

# Shanu Dengre

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## Summary

Customer-facing AI Engineer and Data Scientist with 4+ years of taking ML systems from problem scoping to production. Builds forecasting, computer vision and LLM-based solutions tuned to each customer's business constraints, deployed across 300+ locations. Improved forecast accuracy by 20% and cut food waste by 15%. PhD in Physics.

## Skills

**AI & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, Transformers

**GenAI / LLMs:** LLM APIs, Agentic Workflows, Prompt Engineering, RAG

**Time Series Forecasting:** DARTS, Nixtla, CHRONOS (Foundation Model)

**Computer Vision:** YOLO-family, OpenCV, SAM

**Software & Deployment:** Python, FastAPI, Docker, Kubernetes, Git/GitLab, CI/CD

**Data Engineering & MLOps:** PySpark, MongoDB, Snowflake, SQL, AWS, MLflow, Sentry

**Analytics & Delivery:** Pandas, NumPy, PowerBI, Scrum, Jira, Confluence

**Languages:** English (Fluent), German (A2.2), Hindi (Native)

## Experience

**InterProductTec – Mumbai**

*Nov 2025 – Present*

*Sr. Data Scientist*

**Demand Forecast Project Lead**

- Lead the demand forecast project end to end, from customer scoping to features in production.
- Deployed automated POS integration pipelines for faster customer onboarding to forecast solutions.
- Developed an in-house newsvendor-based ordering optimisation balancing margin against stockout risk, tuning forecast quantiles to each customer's requirements to minimise food waste without losing sales.
- Innovated forecasting solutions improving sales forecast by 20% during holiday/offer seasons.
- Enhanced pipeline observability with Sentry and PowerBI dashboards, achieving negligible downtime while delivering SOTA forecast accuracy across 300+ locations.
- Built LLM-powered agentic workflows (tool-calling agents) to automate internal analytics and reporting.
- Automated customer food waste reports, speeding up delivery and reducing manual effort by 50%.

**PreciTaste – Munich**

*Feb 2022 – Oct 2025*

*AI Engineer and Data Scientist*

**Time Series Forecasting – Sales and Demand**

- Led AI-driven demand forecasting, reducing food waste by 15% and improving crew efficiency by 20%.
- Designed an automated forecasting pipeline using DARTS, cutting retraining time by 50%.
- Reduced query latency by 60% using FastAPI microservices with MongoDB, Docker, and Kubernetes.

**Computer Vision – Object Detection and Segmentation**

- Deployed AI-based food recognition using YOLO-family and PyTorch across 10+ restaurant chains.
- Built automated annotation pipelines with OpenCV and SAM, reducing labeling time by 50%.
- Delivered a POS order-matching system with 95%+ real-time validation accuracy.

## Education

**Data Science & Machine Learning Bootcamp**, Le Wagon Berlin

June 2021 – Sept 2021

Led project: restaurant location recommender (Python, TensorFlow, NLP, GCP, Streamlit)

**PhD (Physics)**, Technical University Dresden

Oct 2015 – Feb 2021

Publication: Phys. Rev. B 103, 064425 (2021)

**BSc–MSc (Physics)**, Indian Institute of Science Education and Research, Pune

Aug 2010 – May 2015

Publication: Phys. Rev. B 95, 054401 (2017)